

Collaborative Learning Journey Begins

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User prompt: Teach me.

Response: Your notes introduce **Uncertainty Quantification (UQ)** and lay out the core ideas behind **Conformal Prediction (CP)**. Let's explore this topic together, and I will ask guiding questions along the way to help step through the foundations.

Based on the outline in your notes, here are three distinct directions we can focus on:

1. **The Core Motivation & Landscape of UQ:** Why point predictions ($\hat{f}(X)$) are insufficient in high-stakes settings (like medicine or law) and how Conformal Prediction compares to Bayesian methods or Quantile Regression.
2. **The Split Conformal Prediction Algorithm:** Walking through the practical steps of training, calibration residuals (R_i), and building the prediction interval $\hat{C}_n(X_{n+1})$.
3. **The Statistical Guarantee & Exchangeability:** Understanding *why* it works mathematically —specifically how ranking and order statistics yield a finite-sample guarantee of $1 - \alpha$ without model assumptions.

Which of these areas would you like to start with?

User prompt: the first 1

Response: Let's explore why uncertainty quantification (UQ) is necessary and how conformal prediction fits into the broader landscape.

A standard machine learning model produces a point prediction: $\hat{f}(X) \rightarrow \hat{Y}$. However, in real-world systems, $\hat{Y} \neq Y$ due to several key sources of error:

-  **Distribution Shift:** The training data distribution differs from real-world deployment ($D_{train} \neq D_{test}$).
-  **Finite Data:** With limited samples, the model cannot capture all possibilities.
-  **Model Approximation Error:** Even the best fitted function $\hat{f}$ rarely matches the true underlying function f^* ($\|\hat{f} - f^*\| \neq 0$).

In high-stakes domains like medicine, finance, or legal decisions, outputting just a single guess $\hat{Y}$ without any measure of doubt can lead to catastrophic failures.

Traditional methods try to quantify this uncertainty, but each comes with tradeoffs:

-  **Bayesian Methods:** Provide full posterior distributions, but require specifying a prior distribution and can carry heavy computational costs.
-  **Quantile Regression:** Directly models specific conditional percentiles, but often suffers from model misspecification and lacks finite-sample guarantees.
-  **Bootstrap:** Resamples the training data, but can be computationally expensive to retrain models repeatedly.

Conformal prediction stands out because it offers **finite-sample validity** and is **model-free**—meaning it wraps around *any* pre-trained model $\hat{f}$ without making distributional assumptions about the data beyond exchangeability.

To see why this distinction matters in practice: suppose a medical diagnostic model predicts a patient has disease risk $\hat{Y} = 0.48$ (just under a 0.50 treatment threshold). What critical piece of information is missing if the doctor only sees that single number $\hat{Y}$?

User prompt: the $\pm$ sign? s

Response: Exactly: the margin of error or spread around that guess (the $\pm$ range).

Without that range, the doctor cannot tell the difference between:

-  High confidence: 0.48 ± 0.01 (safely below the 0.50 treatment threshold)
-  Low confidence: 0.48 ± 0.15 (the true risk could realistically be well above 0.50)

Instead of returning just a point $\hat{Y}$, conformal prediction constructs a **prediction set** or **interval** $\hat{C}_n(X)$ that guarantees with high probability (say, $1 - \alpha = 95\%$) that the true label Y falls inside:

$$P(Y_{n+1} \in \hat{C}_n(X_{n+1})) \geq 1 - \alpha$$

If we set our desired coverage to 95% ($\alpha = 0.05$), what would you expect to happen to the width of that $\pm$ interval if we decide we need an even safer 99% guarantee ($\alpha = 0.01$)?

User prompt: it gets smaller. wait why though/

Response: Think of the interval like casting a net  to catch the true outcome Y :

- If you want to be 95% **sure** the fish is inside your net, you can use a fairly standard net.
- If you now demand to be 99% **sure**—leaving only a tiny 1% chance that the fish escapes—would your net need to be bigger or smaller to guarantee catching it?

If an interval got *smaller*, we would be excluding more possible outcomes, which would actually make it *easier* to miss the true value!

To guarantee higher coverage ($1 - \alpha = 0.99$ instead of 0.95), we have to widen the net .

Now consider the extreme case to see this trade-off: if you wanted a **100% guarantee** with absolute certainty, what kind of interval would you have to output? (Notice the line in your notes: $\hat{C}_n = (-\infty, \infty)$!)

User prompt: right we need infinite to infinite

Response: That is exactly right. An interval of $(-\infty, \infty)$ gives you a 100% guarantee, but it tells the decision-maker virtually nothing useful.

This brings us to the core tension in Uncertainty Quantification:

-  **Coverage:** We want the true label inside our set with high probability ($1 - \alpha$).
-  **Efficiency (Informativeness):** We want the size of the set $|\hat{C}_n|$ to be as small as possible so it remains practical.

Now that we see why point predictions fail and what a valid prediction set aims to achieve, how would you like to proceed?

We can jump into **Part 2: The Split Conformal Prediction Algorithm** (how we actually calculate that finite interval step-by-step using calibration residuals), or dive deeper into the assumptions like **exchangeability**.

User prompt: ok part 2

Response: Let's break down **Split Conformal Prediction** step by step as outlined in your notes.

The goal is to turn any point prediction $\hat{f}(X)$ into a valid interval $[\hat{f}(X) - \hat{q}, \hat{f}(X) + \hat{q}]$. To do this without overfitting, we split our data into two separate sets:

-  **Training Set (D_1):** Used exclusively to fit our predictive model $\hat{f}$.
-  **Calibration Set (D_2):** A set of n_2 unseen data points (X_i, Y_i) used solely to evaluate how wrong the model tends to be.

Step 1: Compute Calibration Residuals

For every point in the calibration set D_2 , we calculate a non-conformity score (the absolute error):

$$R_i = |Y_i - \hat{f}(X_i)| \quad \text{for each } i \in D_2$$

This gives us a list of residual numbers: $\{R_1, R_2, \dots, R_{n_2}\}$.

Step 2: Find the Quantile $\hat{q}$

Next, we sort these residuals from smallest to largest:

$$R_{(1)} \leq R_{(2)} \leq \dots \leq R_{(n_2)}$$

We want to find a cutoff value $\hat{q}$ such that our desired coverage level $1 - \alpha$ is satisfied. The formula from your notes picks a specific order statistic:

$$k = \lceil (1 - \alpha)(n_2 + 1) \rceil$$

$$\hat{q} = R_{(k)}$$

Step 3: Form the Prediction Interval

For any new test input X_{n+1} , our prediction interval is simply:

$$\hat{C}(X_{n+1}) = [\hat{f}(X_{n+1}) - \hat{q}, \hat{f}(X_{n+1}) + \hat{q}]$$

Let's try an example!

Let's test the formula from your notes:

- Suppose our calibration set has $n_2 = 9$ samples.
- We set our desired coverage to 80% ($1 - \alpha = 0.80$).

Using the index formula $k = \lceil (1 - \alpha)(n_2 + 1) \rceil$, which ranked residual $R_{(k)}$ should we pick as our cutoff $\hat{q}$? (Remember, $\lceil \cdot \rceil$ rounds up to the nearest integer).

User prompt: ceiling of $(0.8)(10) = 8$ so we should pick $R(8)$ and so like $\pm R(8)$

Response: Spot on: $k = \lceil 0.80 \times 10 \rceil = 8$, so the cutoff $\hat{q}$ is indeed the 8th smallest residual $R_{(8)}$!

Our prediction interval for any new input X_{n+1} is simply:

$$\hat{C}(X_{n+1}) = [\hat{f}(X_{n+1}) - R_{(8)}, \hat{f}(X_{n+1}) + R_{(8)}]$$

Notice that since $R_{(8)}$ is the 8th value out of 9, exactly 8 out of our 9 calibration points (about 88.9%) have residuals smaller than or equal to $R_{(8)}$.

Now, notice an interesting detail in the formula: Why did we multiply by $n_2 + 1$ instead of just n_2 ?

Think about how many total data points we are actually considering when we include our unseen test point (X_{n+1}, Y_{n+1}) alongside the n_2 calibration points. What do you think?

User prompt: so we have $n_2 + 1$ total points if we include both calibration and the single test point

Response: Exactly. When we include the test point X_{n+1} , there are $n_2 + 1$ total points in play.

This leads directly into **Part 3** from your notes: why this method works mathematically through **exchangeability**.

Consider all $n_2 + 1$ residuals:

$$\{R_1, R_2, \dots, R_{n_2}, R_{n+1}\}$$

Because our data points are drawn independently from the same distribution (they are **exchangeable**), the unseen test residual R_{n+1} is equally likely to fall into any position among the sorted values.

There are $n_2 + 1$ possible ranks for R_{n+1} to end up in:

- It could be the smallest: rank 1
- It could be in the middle: rank 2, 3, $\dots$, n_2
- It could be the largest: rank $n_2 + 1$

Each of these $n_2 + 1$ positions has the exact same probability:

$$P(\text{rank}(R_{n+1}) = j) = \frac{1}{n_2 + 1} \quad \text{for } j \in \{1, 2, \dots, n_2 + 1\}$$

Remember, our prediction succeeds ($Y_{n+1} \in \hat{C}(X_{n+1})$) whenever $R_{n+1} \leq \hat{q}$. In our example with $n_2 = 9$, we set $\hat{q} = R_{(8)}$.

If R_{n+1} can land uniformly anywhere among the 10 positions, what is the exact probability that its rank is ≤ 8 ?

User prompt: I dont understand this phrase: Remember, our prediction succeeds ($Y_{n+1} \in \hat{C}(X_{n+1})$) whenever $R_{n+1} \leq \hat{q}$. In our example with $n_2 = 9$, we set $\hat{q} = R_{(8)}$. but the prob. that rank ≤ 8 is 80%

Response: Let's break down both pieces so the connection becomes clear!

1. Why does $Y_{n+1} \in \hat{C}(X_{n+1})$ mean $R_{n+1} \leq \hat{q}$?

Recall how we built the prediction interval around our guess $\hat{f}(X_{n+1})$:

$$\hat{C}(X_{n+1}) = [\hat{f}(X_{n+1}) - \hat{q}, \hat{f}(X_{n+1}) + \hat{q}]$$

Saying that the true value Y_{n+1} falls inside this interval means:

$$\hat{f}(X_{n+1}) - \hat{q} \leq Y_{n+1} \leq \hat{f}(X_{n+1}) + \hat{q}$$

Subtract $\hat{f}(X_{n+1})$ across the entire inequality:

$$-\hat{q} \leq Y_{n+1} - \hat{f}(X_{n+1}) \leq \hat{q}$$

This is the definition of absolute value:

$$|Y_{n+1} - \hat{f}(X_{n+1})| \leq \hat{q}$$

Since the residual error is defined as $R_{n+1} = |Y_{n+1} - \hat{f}(X_{n+1})|$, the statement simplifies directly to:

$$R_{n+1} \leq \hat{q}$$

The interval catches the true outcome as long as the test error is no larger than our cutoff $\hat{q}$.

2. How does this connect to your 80% calculation?

Your calculation is spot on:

$$\frac{8}{10} = 0.80 = 80\%$$

Because all 10 ranks are equally likely due to exchangeability, the chance that the new test residual lands in rank 1, 2, ..., or 8 is exactly $\frac{8}{10} = 80\%$.

Since we chose $\hat{q} = R_{(8)}$, any test error with rank ≤ 8 is $\leq \hat{q}$. Therefore:

$$P(Y_{n+1} \in \hat{C}(X_{n+1})) = P(R_{n+1} \leq R_{(8)}) = 80\%$$

This achieves the target coverage of $1 - \alpha = 0.80$ without making any assumptions about what the true distribution looks like!

To see what happens when the math doesn't divide evenly: what if our desired coverage was 75% ($1 - \alpha = 0.75$) with the same $n_2 = 9$ points?

Using the formula $k = \lceil (1 - \alpha)(n_2 + 1) \rceil$, which rank $R_{(k)}$ would we choose, and what would our actual coverage probability be?

User prompt: (0.75)(10) ceiling is 8. And so, we choose $R(8)$. The coverage prob. is the same then?

Response: Let's check the arithmetic:

$$0.75 \times 10 = 7.5$$

Taking the ceiling (rounding up to the nearest integer) gives:

$$\lceil 7.5 \rceil = 8$$

Because $k = 8$, we select the exact same residual $R_{(8)}$ as our threshold $\hat{q}$!

Our actual coverage probability is still:

$$P(\text{rank}(R_{n+1}) \leq 8) = \frac{8}{10} = 80\%$$

Notice how the mathematical guarantee holds:

$$P(Y_{n+1} \in \hat{C}(X_{n+1})) = 0.80 \geq 0.75$$

The guarantee says coverage will be **at least** $1 - \alpha$ ($80\% \geq 75\%$). Because we only have a discrete set of 10 positions, the ceiling function guarantees we never under-cover, even if it makes our interval slightly more conservative than strictly requested.

To see what happens as the calibration set grows, what would happen to that gap between the requested coverage and the actual coverage if we had $n_2 = 999$ calibration points instead of $n_2 = 9$?

User prompt: then it will be more finegrained, so like (0.75)(1000) is 750 so $R(750)$ while the 0.8 is $R(800)$

Response: That is exactly it.

With $n_2 = 999$, we have $n_2 + 1 = 1000$ total positions:

- 🎯 For $1 - \alpha = 0.75$: $k = \lceil 0.75 \times 1000 \rceil = 750$, giving an exact coverage of $\frac{750}{1000} = 75.0\%$.
- 🎯 For $1 - \alpha = 0.80$: $k = \lceil 0.80 \times 1000 \rceil = 800$, giving an exact coverage of $\frac{800}{1000} = 80.0\%$.

As our calibration set grows, the empirical quantiles become much more fine-grained, allowing us to hit our target coverage level almost exactly without having to "over-cover" due to rounding up.

There is also an upper-bound theorem in conformal prediction:

$$1 - \alpha \leq P(Y_{n+1} \in \hat{C}(X_{n+1})) \leq 1 - \alpha + \frac{1}{n_2 + 1}$$

Notice that as $n_2 \rightarrow \infty$, the term $\frac{1}{n_2 + 1}$ shrinks to 0, meaning the actual coverage tightens right onto the requested $1 - \alpha$.

A limitation of this basic method

Look closely at the interval we constructed:

$$\hat{C}(X) = [\hat{f}(X) - \hat{q}, \hat{f}(X) + \hat{q}]$$

Notice that the width of this interval is always $2\hat{q}$, regardless of what input X we feed in.

In a real-world scenario (like predicting house prices or patient recovery times), why might having the exact same interval width $2\hat{q}$ for **every single input** X be undesirable?

User prompt: why? sometimes we want it to be larger or narrower when we are more sure? im not sure

Response: You hit the intuition right on the head!

Uncertainty is rarely uniform across the entire input space:

- 🎯 **High certainty:** For a typical, common case (e.g., predicting the price of a standard 3-bedroom suburban home, or a patient with textbook symptoms), our model is usually quite accurate, so a narrow interval is appropriate.
- ? **Low certainty:** For a rare, unusual case (e.g., a custom luxury mansion, or an elderly patient with multiple complex conditions), the model's predictions are naturally noisier and far more uncertain. Here, we *want* a wider interval to capture that higher spread.

Because standard split conformal prediction calculates a single global cutoff $\hat{q}$, it forces every prediction interval to have the exact same fixed width ($2\hat{q}$):

$$\hat{C}(X) = [\hat{f}(X) - \hat{q}, \hat{f}(X) + \hat{q}]$$

This means:

- ⚠ It tends to be **unnecessarily wide** for easy inputs where the model is very sure.
- ⚠ It tends to be **dangerously narrow** for difficult, rare inputs where uncertainty is large.

While it guarantees marginal coverage overall (on average across all inputs), it can fail to provide good *conditional* coverage for specific types of inputs.

To adapt the interval width to how certain we are for a given input X , how might we adjust our residual score $R_i = |Y_i - \hat{f}(X_i)|$ so that hard inputs naturally get wider intervals than easy ones? (Think about dividing by something that estimates local variability!)

User prompt: i dont know

Response: To get adaptive intervals, we normalize the raw error by a local measure of spread or uncertainty, often denoted as $\hat{\sigma}(X)$:

$$R_i = \frac{|Y_i - \hat{f}(X_i)|}{\hat{\sigma}(X_i)}$$

Here, $\hat{\sigma}(X)$ is a separate estimate of how noisy or difficult predicting around X is (for example, the standard deviation or an error model).

When we unwrap this scaled score to form the prediction interval, we get:

$$\hat{C}(X_{n+1}) = [\hat{f}(X_{n+1}) - \hat{q} \cdot \hat{\sigma}(X_{n+1}), \hat{f}(X_{n+1}) + \hat{q} \cdot \hat{\sigma}(X_{n+1})]$$

Notice the width of the interval is now $2\hat{q} \cdot \hat{\sigma}(X_{n+1})$.

Looking at that interval width:

- 📏 When a test case is easy and familiar, $\hat{\sigma}(X_{n+1})$ will be small, making the interval naturally narrow.
- 📏 When a test case is unusual or difficult, $\hat{\sigma}(X_{n+1})$ will be large, making the interval wider.

This approach is called **Locally Weighted or Normalized Conformal Prediction**.

Because the calibration math relies only on ranking the relative scores R_i , the exact coverage guarantee $P(Y_{n+1} \in \hat{C}(X_{n+1})) \geq 1 - \alpha$ still holds—even if our estimate $\hat{\sigma}(X)$ is imperfect!

Now that we have covered regression and adaptive intervals, how would you like to proceed? We could look at:

1. 🟡 **Classification / Discrete Labels:** How conformal prediction outputs a set of possible labels (like {cat, dog}) instead of numbers.
2. ⚠️ **Distribution Shift & Assumptions:** What happens when the test data does not follow the calibration distribution (breaking exchangeability).

User prompt: why this form: $\hat{C}(X_{n+1}) = \left[\hat{f}(X_{n+1}) - \hat{q} \cdot \hat{\sigma}(X_{n+1}), \hat{f}(X_{n+1}) + \hat{q} \cdot \hat{\sigma}(X_{n+1}) \right]$ I also don't understand this.. Because the calibration math relies only on ranking the relative scores R_i , the exact coverage guarantee $P(Y_{n+1} \in \hat{C}(X_{n+1})) \geq 1 - \alpha$ still holds—even if our estimate $\hat{\sigma}(X)$ is imperfect! like we are multiplying and making is smaller and larger. how can this still hold.

Response: Let's trace the algebra step by step to see why that formula appears and why the guarantee never breaks, even if $\hat{\sigma}(X)$ is a terrible estimate!

1. Where does the formula come from?

Remember how we set up our condition for success:

$$R_{n+1} \leq \hat{q}$$

In this adaptive version, our non-conformity score was defined by dividing by $\hat{\sigma}(X)$:

$$R_{n+1} = \frac{|Y_{n+1} - \hat{f}(X_{n+1})|}{\hat{\sigma}(X_{n+1})}$$

Now plug this definition directly into $R_{n+1} \leq \hat{q}$:

$$\frac{|Y_{n+1} - \hat{f}(X_{n+1})|}{\hat{\sigma}(X_{n+1})} \leq \hat{q}$$

Assuming $\hat{\sigma}(X_{n+1}) > 0$, multiply both sides by $\hat{\sigma}(X_{n+1})$:

$$|Y_{n+1} - \hat{f}(X_{n+1})| \leq \hat{q} \cdot \hat{\sigma}(X_{n+1})$$

Now unpack the absolute value:

$$-\hat{q} \cdot \hat{\sigma}(X_{n+1}) \leq Y_{n+1} - \hat{f}(X_{n+1}) \leq \hat{q} \cdot \hat{\sigma}(X_{n+1})$$

Add $\hat{f}(X_{n+1})$ to all three parts:

$$\hat{f}(X_{n+1}) - \hat{q} \cdot \hat{\sigma}(X_{n+1}) \leq Y_{n+1} \leq \hat{f}(X_{n+1}) + \hat{q} \cdot \hat{\sigma}(X_{n+1})$$

This inequality defines the interval!

$$\hat{C}(X_{n+1}) = \left[\hat{f}(X_{n+1}) - \hat{q} \cdot \hat{\sigma}(X_{n+1}), \hat{f}(X_{n+1}) + \hat{q} \cdot \hat{\sigma}(X_{n+1}) \right]$$

2. Why does the coverage guarantee still hold if $\hat{\sigma}$ is imperfect?

Here is the key insight: **we apply the exact same function $\hat{\sigma}(X)$ to both the calibration points and the test point.**

Let's look at the ranks again:

- Calibration scores: $R_i = \frac{|Y_i - \hat{f}(X_i)|}{\hat{\sigma}(X_i)}$ for $i = 1, \dots, n_2$
- Test score: $R_{n+1} = \frac{|Y_{n+1} - \hat{f}(X_{n+1})|}{\hat{\sigma}(X_{n+1})}$

Because the pairs $(X_1, Y_1), \dots, (X_{n+1}, Y_{n+1})$ are exchangeable (i.i.d.), the scores $R_1, R_2, \dots, R_{n+1}$ are **also exchangeable**.

It does not matter how weird or messy the score function is:

- R_{n+1} is still just a random number drawn from the exact same pool as the other n_2 numbers.
- The rank of R_{n+1} among all $n_2 + 1$ scores is **still perfectly uniform**: each position from 1 to $n_2 + 1$ has probability $\frac{1}{n_2 + 1}$.

If $\hat{\sigma}(X)$ is wildly off—say it accidentally predicts values that are 10 times too small across the board—what will happen to the calibration scores R_i ? And what will that do to the threshold $\hat{q}$?

User prompt: what?

Response: Let's look at a quick concrete number example to see how the math self-corrects .

Suppose our true error for every point is 10:

- Without scaling: $|Y - \hat{f}(X)| = 10$.
- So our cutoff is simply $\hat{q} = 10$.
- Our interval width is $\pm \hat{q} = \pm 10$.

Now, suppose we use a terrible model $\hat{\sigma}(X)$ that estimates uncertainty as 2 everywhere instead of 1:

1.  **Calibration Scores get divided:** Each calibration score becomes:

$$R_i = \frac{10}{2} = 5$$

So our new quantile cutoff is $\hat{q} = 5$.

2.  **The Prediction Interval multiplies it right back:** The formula multiplies the cutoff by $\hat{\sigma}$:

$$\hat{q} \cdot \hat{\sigma}(X) = 5 \cdot 2 = 10$$

Notice what happened:

- Dividing by $\hat{\sigma}$ in calibration made the threshold $\hat{q}$ half as big ($10 \rightarrow 5$).
- Multiplying by $\hat{\sigma}$ when creating the interval doubled it right back ($5 \times 2 = 10$).

The scaling completely cancels out in the overall scale!

Why does this guarantee coverage?

Coverage only depends on whether:

$$R_{n+1} \leq \hat{q}$$

Since we applied the exact same formula to calculate the calibration scores $R_1, \dots, R_n$ and the test score R_{n+1} , their relative order (ranking) does not change.

If R_{n+1} was smaller than 80% of the calibration points before scaling, it will still be smaller than 80% of them after scaling!

Does seeing how the division and multiplication cancel out make the intuition clearer?

User prompt: I think you might need to give me a side by side comparison of the without the multipleication and with it.

Response: Here is a side-by-side comparison showing how standard conformal prediction differs from the adaptive (scaled) version:

Feature	Standard Conformal Prediction	Locally Weighted (Adaptive) Conformal Prediction
Calibration Score (R_i)	$R_i = \ Y_i - \hat{f}(X_i)\ $	$R_i = \frac{\ Y_i - \hat{f}(X_i)\ }{\hat{\sigma}(X_i)}$
Threshold ($\hat{q}$)	k -th smallest value of raw errors	k -th smallest value of <i>scaled</i> errors
Prediction Interval	$[\hat{f}(X) - \hat{q}, \hat{f}(X) + \hat{q}]$	$[\hat{f}(X) - \hat{q}\hat{\sigma}(X), \hat{f}(X) + \hat{q}\hat{\sigma}(X)]$
Interval Width	Constant: $2\hat{q}$ for every input X	Dynamic: $2\hat{q}\hat{\sigma}(X)$, changes with each input X
Easy vs. Hard Inputs	Gives the exact same band size to both simple and difficult cases.	Shrinks when $\hat{\sigma}(X)$ is small (easy cases); expands when $\hat{\sigma}(X)$ is large (hard cases).
Coverage Guarantee	$\geq 1 - \alpha$	$\geq 1 - \alpha$

What changes and what stays the same?

- 🔄 **What stays the same:** In both methods, the math only counts the **rank** of the test score among the calibration scores. Because every data point goes through the exact same scoring formula, the probability that the new score falls within the top $1 - \alpha$ cutoff remains mathematically guaranteed at $\geq 1 - \alpha$.
- 🔧 **What changes:** By dividing by $\hat{\sigma}(X)$ when computing scores and multiplying by $\hat{\sigma}(X)$ when drawing the interval, we reshape the boundaries so that uncertain inputs receive wider bands and confident inputs receive narrower bands.

Looking at this comparison, what do you think would happen if our uncertainty estimator $\hat{\sigma}(X)$ was just a constant number (like $\hat{\sigma}(X) = 3$ for all X)?

User prompt: its just everything divided by 3 so its fair?

Response: That is spot on.

Because every calibration score is divided by 3, the cutoff value $\hat{q}$ simply becomes 3 times smaller. Then, when forming the interval, we multiply that cutoff by 3 again:

$$\hat{q}_{\text{scaled}} \cdot \hat{\sigma}(X) = \left(\frac{\hat{q}_{\text{standard}}}{3} \right) \cdot 3 = \hat{q}_{\text{standard}}$$

The 3 completely cancels out, collapsing right back into the standard, fixed-width conformal prediction!

This shows why $\hat{\sigma}(X)$ has to vary across inputs if we want genuinely adaptive bands. It needs to reflect how uncertainty changes depending on the features of X .

We have covered regression (continuous outcomes) and how intervals are formed. Where would you like to explore next from your lecture notes?

1. 🍷 **Classification:** How conformal prediction constructs **discrete prediction sets** (e.g., predicting {cat, dog} instead of a numerical range).
2. ⚠️ **Distribution Shift:** What happens when the test data violates the exchangeability assumption (like covariate shift).

User prompt: but what if its not a constant? that doesnt seem fair then?

Response: It feels counterintuitive at first because different points get scaled by different amounts. But here is why the probability guarantee remains mathematically "fair" and valid:

1. The Rule is Fixed Beforehand 🗝️

Remember that the scaling model $\hat{\sigma}(X)$ is trained **only on** D_1 (the training data). Once training is done, $\hat{\sigma}(X)$ is completely frozen. It is simply a fixed mathematical formula.

2. Every Pair Gets the Exact Same Treatment 🤝

Because $(X_1, Y_1), \dots, (X_{n_2}, Y_{n_2})$ and the new test point (X_{n+1}, Y_{n+1}) are drawn independently from the exact same data distribution (they are **exchangeable**), each point goes through the exact same scoring machine:

$$R = \frac{|Y - \hat{f}(X)|}{\hat{\sigma}(X)}$$

Since every (X_i, Y_i) pair comes from the same underlying distribution and gets processed by the exact same rule, the resulting numbers:

$$\{R_1, R_2, \dots, R_{n_2}, R_{n+1}\}$$

are **still exchangeable** random variables!

3. Uniform Ranks Still Hold 🎮

Because all $n_2 + 1$ scores are exchangeable, the test score R_{n+1} has no special bias. It is just as likely to be ranked 1st, 5th, or last among the scores:

$$P(\text{rank}(R_{n+1}) \leq k) = \frac{k}{n_2 + 1}$$

And by definition:

$$\text{rank}(R_{n+1}) \leq k \iff R_{n+1} \leq \hat{q} \iff Y_{n+1} \in \hat{C}(X_{n+1})$$

The math does not care what numbers R represents—as long as the scores are exchangeable, the ranking guarantee holds!

To test this: what would happen if we peeked at the calibration set D_2 and used it to fit $\hat{\sigma}(X)$ instead of keeping $\hat{\sigma}$ fixed from D_1 ? Would the calibration scores and the test score still be treated identically?

User prompt: ok so basically we are saying this sigma thing is allowed because the data are exchangeable? and ur question.. humm yes?

Response: For your first thought: **yes**, exactly. As long as the rule $\hat{\sigma}(X)$ is fixed beforehand, applying it to exchangeable data points yields exchangeable scores. The math works because the symmetry between the points is completely preserved.

Now for the question about peeking at D_2 : think about what happens when a model is trained on specific data.

- If we use the calibration set D_2 to fit $\hat{\sigma}(X)$, $\hat{\sigma}$ gets to learn all the specific patterns and quirks of the calibration points.
- But the unseen test point (X_{n+1}, Y_{n+1}) was **not** seen during that training.

When a model is evaluated on data it was trained on versus data it has never seen, what usually happens to the errors on the training data compared to the new test data?

User prompt: it will overfit to the calibration set... so the unseen point is no longer exchangeable? there is bias? but does bias actually violate exchangeability?

Response: You've hit on the exact reason why peeking breaks the system!

Here is why overfitting directly violates **exchangeability**:

-  **Exchangeability requires symmetry:** Mathematically, variables $R_1, R_2, \dots, R_{n+1}$ are exchangeable if their joint probability distribution doesn't change when you swap any of their positions. Every point must play the exact same statistical role.
-  **Overfitting breaks the symmetry:** If you use the calibration set D_2 to fit $\hat{\sigma}(X)$, $\hat{\sigma}$ specifically adapts to minimize or tailor errors on those exact calibration points.
 - The calibration residuals $R_1, \dots, R_{n_2}$ will tend to be **artificially small**.
 - The test residual R_{n+1} was never seen by the model, so it won't receive that special treatment and will tend to be **larger**.

Because R_{n+1} is systematically drawn from a distribution with larger errors than the calibration scores, the positions are no longer symmetric.

R_{n+1} is no longer equally likely to land in any of the $n_2 + 1$ slots—it is much more likely to land near the top (the largest ranks).

Since R_{n+1} is systematically pushed into the higher ranks, what do you think happens to our coverage guarantee ($P(R_{n+1} \leq \hat{q}) \geq 1 - \alpha$)? Will our intervals cover the true test value as often as promised?

User prompt: no? since we have this calibration dataset that yield the $\hat{q}$ but the model is already trained on it so the $\hat{q}$ is likely very small, and then apply that to unseen data is gonna be bad

Response: Precisely.

Because the model was trained on the calibration points, it already knows them well, making the residuals $R_1, \dots, R_{n_2}$ artificially small.

-  **The threshold $\hat{q}$ shrinks:** Since the calibration errors are unrealistically optimistic, the cutoff $\hat{q}$ will be chosen far too small.
-  **The intervals become too tight:** When you build the interval $[\hat{f}(X) - \hat{q}, \hat{f}(X) + \hat{q}]$, it will be way too narrow to handle genuine, unseen test data.
-  **Coverage drops:** Instead of covering the true label 90% of the time, the true coverage might plummet to 70% or even lower.

This is why the strict separation in your lecture notes ($D_1 \cap D_2 = \emptyset$) is so essential. The calibration set D_2 must remain completely untouched during model training so that its errors mirror what the model will actually encounter in the wild.

Now that we have covered:

1. 💡 Why uncertainty quantification matters
2. ⚙️ How split conformal prediction calculates the interval step-by-step
3. 🎲 Why exchangeability and data-splitting provide the mathematical guarantee

We can explore how to handle **discrete classification** (e.g., getting a prediction set of labels like {cat, dog}), or look at what happens when the real world violates exchangeability via **distribution shift** (covariate shift). Which direction sounds most interesting?

User prompt: ok discrete classificaiton.

Response: In classification, the outcome Y is not a continuous number, but a categorical label, such as {cat, dog, fox, wolf}.

Instead of an interval $[a, b]$, our prediction set $\hat{C}(X_{n+1})$ is a **set of candidate labels**:

$$P(Y_{n+1} \in \hat{C}(X_{n+1})) \geq 1 - \alpha$$

Here is how we adapt the conformal prediction recipe:

Step 1: Model Output & Non-Conformity Score

A standard classification model typically outputs a predicted probability for each class y :

$$\hat{\pi}_y(X) = \hat{P}(Y = y \mid X)$$

A natural choice for a **non-conformity score** (how surprising or unexpected the true label is) is:

$$R_i = 1 - \hat{\pi}_{Y_i}(X_i)$$

- 🟢 If the model assigned a high probability to the correct class (e.g., 0.90), the score is small: $1 - 0.90 = 0.10$.
- 🟡 If the model assigned a low probability to the correct class (e.g., 0.05), the score is large: $1 - 0.05 = 0.95$.

Step 2: Calibrate

Just like in regression, we compute R_i on each point in the calibration set D_2 and find the cutoff quantile $\hat{q} = R_{(k)}$.

Step 3: Form the Prediction Set

For a new input X_{n+1} , we test **every possible class label** y . We include label y in our set if its non-conformity score is small enough:

$$\hat{C}(X_{n+1}) = \{y : 1 - \hat{\pi}_y(X_{n+1}) \leq \hat{q}\}$$

Which is equivalent to:

$$\hat{C}(X_{n+1}) = \{y : \hat{\pi}_y(X_{n+1}) \geq 1 - \hat{q}\}$$

To see how this works: suppose our calibration cutoff is $\hat{q} = 0.70$, so our threshold probability is $1 - \hat{q} = 0.30$.

If our model predicts the following probabilities for an image:

- 🐱 Cat: 0.65
- 🐶 Dog: 0.25
- 🦊 Fox: 0.10

Which labels would be included in the prediction set $\hat{C}(X_{n+1})$?

User prompt: cat? since 63 > 30?

Response: Cat is definitely included because $0.65 \geq 0.30$.

Looking closely at the other probabilities:

- 🐶 Dog: 0.25
- 🦊 Fox: 0.10

Both 0.25 and 0.10 are strictly below the 0.30 threshold, so they are excluded. Our prediction set is simply:

$$\hat{C}(X_{n+1}) = \{\text{Cat}\}$$

Because the model was confident in "Cat", the prediction set contains just a single class.

Now, consider a more ambiguous case: suppose an image is blurry, and the model outputs:

- 🐱 Cat: 0.45
- 🐶 Dog: 0.35

- 🦊 Fox: 0.20

Using the exact same threshold ($1 - \hat{q} = 0.30$), which labels would make it into the prediction set $\hat{C}(X_{n+1})$ this time?

User prompt: both cat and dog? cuz 45 and 35 both ≥ 30

Response: Both **Cat** and **Dog** meet the condition because:

- 🐱 Cat: $0.45 \geq 0.30$
- 🐶 Dog: $0.35 \geq 0.30$
- 🦊 Fox: $0.20 < 0.30$ (excluded)

So the prediction set is:

$$\hat{C}(X_{n+1}) = \{\text{Cat}, \text{Dog}\}$$

Notice how naturally this adapts to model uncertainty:

- 🎯 When the image was clear, the set had only 1 **label** ($\{\text{Cat}\}$).
- 🖼️ When the image was ambiguous, the set grew to 2 **labels** ($\{\text{Cat}, \text{Dog}\}$) to preserve the coverage guarantee without forcing an arbitrary guess.

An Edge Case: High Uncertainty

What would happen if an input was completely uninformative or noisy, resulting in nearly uniform probabilities across all three classes:

- 🐱 Cat: 0.34
- 🐶 Dog: 0.33
- 🦊 Fox: 0.33

With the same threshold (0.30), what would the prediction set look like this time?

User prompt: all 3 outputed?

Response: All three labels are included:

$$\hat{C}(X_{n+1}) = \{\text{Cat}, \text{Dog}, \text{Fox}\}$$

Because every class has a predicted probability ≥ 0.30 , none can be safely eliminated while maintaining the coverage guarantee.

This demonstrates how the set size reflects confidence:

-  **High confidence:** $\{\text{Cat}\}$ (size 1)
-  **Ambiguity:** $\{\text{Cat}, \text{Dog}\}$ (size 2)
-  **Complete uncertainty:** $\{\text{Cat}, \text{Dog}, \text{Fox}\}$ (size 3)

A Curious Edge Case: The Empty Set 🛑

Consider what happens if an image is completely out-of-distribution (e.g., static noise) and we had 10 classes, with the model spreading probability evenly so each class gets only 0.10.

With that same cutoff threshold of 0.30, how many labels would satisfy $\hat{\pi}_y(X_{n+1}) \geq 0.30$? What would the prediction set $\hat{C}(X_{n+1})$ contain?

User prompt: none? the prediction set is 0

Response: Exactly: none of the labels meet the threshold, so the prediction set is the **empty set** $\emptyset$ (size 0)!

While returning an empty set seems odd, it provides useful information:

-  **Anomaly Detection:** An empty set signals that the model finds *every* candidate class unconvincing. It serves as a warning that the input might be completely out-of-distribution, corrupted, or not belong to any of the known classes.
-  **Coverage Meaning:** Since the true label cannot be in an empty set, an empty set is automatically an error ($Y_{n+1} \notin \emptyset$). The math allows this to happen up to α percent of the time.

Tying it all together

We have covered three core pillars from your notes:

1.  **Why UQ Matters:** Finite data, approximation error, and distribution shifts make point predictions risky in high-stakes fields.
2.  **The Conformal Framework:** Splitting data into training D_1 and calibration D_2 guarantees validity because exchangeable samples produce uniformly distributed ranks.

3.  **Continuous vs. Discrete:** We can construct numeric intervals $[\hat{f}(X) \pm \hat{q}]$ for regression or dynamic sets of labels for classification.

From the initial list in your notes, we can explore:

-  **Distribution Shift / Covariate Shift:** What happens to our guarantees when $D_{\text{train}} \neq D_{\text{test}}$?
-  **Online Learning / Generative Models / LLMs:** How conformal prediction is applied to modern models.

Where would you like to direct our focus next?
